

CAT II & III: 3rd YEAR SECOND SEMESTER, BSc MECHATRONIC ENGINEERING
EMT 3205 MEASUREMENT AND INSTRUMENTATION

Q1. Name four types of AC bridges and state the primary measurement quantity each is used to measure. [4 marks]

Q2. An industrial plant uses several analog instruments to measure electrical and mechanical parameters. The engineer needs to select the appropriate instruments for various tasks based on the following requirements.

- I. A **direct measurement** of the electrical current is needed for monitoring the plant's power supply. The instrument must provide a reading on a calibrated scale by the deflection of a pointer. **Which type of instrument is best suited for this application, and why?**
[3 marks]
- II. A **precision measurement** of the electrical resistance is required to detect slight variations in material conductivity. The instrument should provide accurate results by balancing the unknown resistance with a known standard, showing zero when the values are equal. **Which type of instrument would be most suitable for this application, and why?** [3 marks]
- III. The plant engineer needs a **general-purpose instrument** to measure the temperature of a machine part. The instrument should be simple and inexpensive, providing a reliable reading of temperature without needing extreme precision. **Which instrument type would you recommend for this task, and why?** [3 marks]
- IV. A **comparison-based measurement** is required to measure an unknown voltage by comparing it with a known standard voltage. The instrument should display the difference between the unknown and standard voltage. **Which type of instrument would be ideal for this application, and why?** [3 marks]

When done
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Q3. List the two main types of instrument transformers and state the primary purpose of each. [4 marks] ^{CT}_{VT}

Q4. Name and briefly explain four types of remote sensing technologies and provide one application example for each. [12 marks] ^{Lidar}_{Radar} ^{Acoustic}_{TIR} ^{Re}

Q5 A mechatronic engineer is designing a vibration monitoring system for a robotic arm in a manufacturing cell. The vibration sensor outputs a weak analog signal of 8 mV, but the data acquisition (DAQ) system requires an input between 0.8V and 2.0V. The sensor signal also contains high-frequency electrical noise from nearby welding machines, and the robot operates in an electrically noisy environment where voltage spikes may occur.

- I. Determine the amplifier gain needed to raise the 8-mV signal to 0.8 V minimum. [2 marks] $G = \frac{V_{out}}{V_{in}}$ ¹⁰⁰
- II. Select a suitable filter type and cutoff frequency to remove the high-frequency noise if the useful vibration frequencies are 0–200 Hz. [2 marks] ^{0.8}_{2.0} ^{low pass} ^{cut 200 Hz}
- III. Calculate the minimum isolation resistance required to limit current to 0.5 mA during a possible 300 V spike. [3 marks]
- IV, Explain why linearization is NOT required for this sensor and justify your answer. [3 marks]

Q6. Define calibration and explain its four benefits in instrumentation. [10 marks] ^{Accuracy}_{Compliance} ^{Standard}_{Praxis}

Q7. State and explain four main applications of microprocessors in instrumentation systems. [8 marks] ^{ATE}_{Oscilloscope} ^{Temperature control} ^{Automated Testing}

^{Low}_{Pass} ^{filter}